Table 8: **Ensemble Adversarial Training on MNIST.** For black-box robustness, we report the maximum and average error rate over a suite of 12 attacks, comprised of the FGSM, I-FGSM and PGD (Madry et al., 2017) attacks applied to models A,B,C and D. We use $\epsilon = 16$ in all cases. For each model architecture, we mark the models tied for best (at a 95% confidence level) in bold.

	Clean	**FGSM**	**Max. Black Box**	**Avg. Black Box**
A_{adv}	**0.8**	**2.2**	10.8	7.7
$A_{adv\text{-}ens}$	**0.8**	7.0	**6.6**	5.2
$A_{adv\text{-}ens3}$	**0.7**	5.4	**6.5**	**4.3**
B_{adv}	**0.8**	**11.6**	8.9	**5.5**
$B_{adv\text{-}ens}$	**0.7**	**10.5**	**6.8**	**5.3**
$B_{adv\text{-}ens3}$	**0.8**	14.0	8.8	**5.1**
C_{adv}	**1.0**	3.7	29.3	18.7
$C_{adv\text{-}ens}$	**1.3**	**1.9**	17.2	10.7
$C_{adv\text{-}ens3}$	**1.4**	3.6	**14.5**	**8.4**
D_{adv}	**2.6**	25.5	32.5	23.5
$D_{adv\text{-}ens}$	**2.6**	**21.5**	38.6	28.0
$D_{adv\text{-}ens3}$	**2.6**	29.4	**29.8**	**15.6**

The attacks we consider are the FGSM, I-FGSM and the PGD attack from Madry et al. (2017) with the loss function from Carlini & Wagner (2017a)), all with $\epsilon = 0.3$. The results appear in Table 8. For each model, we report the worst-case and average-case error rate over all black-box attacks.

Ensemble Adversarial Training significantly increases robustness to black-box attacks, except for architecture B, which we previously found to not suffer from the same overfitting phenomenon that affects the other adversarially trained networks. Nevertheless, model $B_{adv\text{-}ens}$ achieves slightly better robustness to white-box and black-box attacks than B_{adv}. In the majority of cases, we find that using a single pre-trained model produces good results, but that the extra diversity of including three pre-trained models can sometimes increase robustness even further. Our experiments confirm our conjecture that robustness to black-box attacks generalizes across models. Indeed, we find that when training with three external models, we attain very good robustness against attacks initiated from models with the same architecture (as evidenced by the average error on our attack suite), but also increased robustness to attacks initiated from the fourth holdout model

D TRANSFERABILITY OF RANDOMIZED SINGLE-STEP PERTURBATIONS.

In Section 4.1, we introduced the R+Step-LL attack, an extension of the Step-LL method that prepends the attack with a small random perturbation. In Table 9, we evaluate the transferability of R+Step-LL adversarial examples on ImageNet. We find that the randomized variant produces perturbations that transfer at a much lower rate (see Table 1 for the deterministic variant).

Table 9: **Error rates (in %) of randomized single-step attacks transferred between models on ImageNet.** We use R+Step-LL with $\epsilon = 16/256, \alpha = \epsilon/2$ for 10,000 random test set samples. The white-box attack always outperforms black-box attacks.

	Source						**Source**				
Target	v4	v3	$v3_{adv}$	IRv2	$IRv2_{adv}$	**Target**	v4	v3	$v3_{adv}$	IRv2	$IRv2_{adv}$
v4	**70.5**	37.2	23.2	34.0	24.6	v4	**42.8**	14.3	6.3	11.9	6.9
v3	42.6	**80.0**	26.7	38.5	27.6	v3	18.0	**57.1**	8.0	15.6	8.6
$v3_{adv}$	31.4	30.7	**64.8**	30.4	34.0	$v3_{adv}$	10.7	10.4	**37.1**	10.1	12.9
IRv2	36.2	35.7	23.0	**56.3**	24.6	IRv2	12.8	13.6	6.1	**29.3**	7.0
$IRv2_{adv}$	26.8	26.3	25.2	26.9	**37.5**	$IRv2_{adv}$	8.0	8.0	7.7	8.3	**15.0**
	Top 1						**Top 5**				

E GRADIENT ALIGNED ADVERSARIAL SUBSPACES FOR THE ℓ_∞ NORM

Tramèr et al. (2017) consider the following task for a given model h: for a (correctly classified) point x, find k orthogonal vectors $\{r_1, \ldots, r_k\}$ such that $\|r_i\|_2 \leq \epsilon$ and all the $x + r_i$ are adversarial (i.e., $\arg\max h(x + r_i) \neq y_{\text{true}}$). By linearizing the model's loss function, this reduces to finding k orthogonal vectors r_i that are maximally aligned with the model's gradient $g = \nabla_x L(h(x), y_{\text{true}})$. Tramèr et al. (2017) left a construction for the ℓ_∞ norm as an open problem.

We provide an optimal construction for the ℓ_∞ norm, based on *Regular Hadamard Matrices* (Colbourn, 2010). Given the ℓ_∞ constraint, we find orthogonal vectors r_i that are maximally aligned with the *signed* gradient, $\texttt{sign}(g)$. We first prove an analog of (Tramèr et al., 2017, Lemma 1).

Lemma 6. *Let $v \in \{-1, 1\}^d$ and $\alpha \in (0, 1)$. Suppose there are k orthogonal vectors $r_1, \ldots r_n \in \{-1, 1\}^d$ satisfying $v^\top r_i \geq \alpha \cdot d$. Then $\alpha \leq k^{-\frac{1}{2}}$.*

Proof. Let $\hat{r_i} = \frac{r_i}{\|r_i\|_2} = \frac{r_i}{\sqrt{d}}$. Then, we have

$$d = \|v\|_2^2 \geq \sum_{i=1}^{k} |v^\top \hat{r_i}|^2 = d^{-1} \sum_{i=1}^{k} |v^\top r_i|^2 \geq d^{-1} \cdot k \cdot (\alpha \cdot d)^2 = k \cdot \alpha^2 \cdot d \,, \tag{12}$$

from which we obtain $\alpha \leq k^{-\frac{1}{2}}$. □

This result bounds the number of orthogonal perturbations we can expect to find, for a given alignment with the signed gradient. As a warm-up consider the following trivial construction of k orthogonal vectors in $\{-1, 1\}^d$ that are "somewhat" aligned with $\texttt{sign}(g)$. We split $\texttt{sign}(g)$ into k "chunks" of size $\frac{d}{k}$ and define r_i to be the vector that is equal to $\texttt{sign}(g)$ in the ith chunk and zero otherwise. We obtain $\texttt{sign}(g)^\top r_i = \frac{d}{k}$, a factor $\sqrt{k}$ worse than the the bound in Lemma 6.

We now provide a construction that meets this upper bound. We make use of Regular Hadamard Matrices of order k (Colbourn, 2010). These are square matrices H_k such that: (1) all entries of H_k are in $\{-1, 1\}^k$; (2) the rows of H_k are mutually orthogonal; (3) All row sums are equal to $\sqrt{k}$.

The order of a Regular Hadamard Matrix is of the form $4u^2$ for an integer u. We use known constructions for $k \in \{4, 16, 36, 64, 100\}$.

Lemma 7. *Let $g \in \mathbb{R}^d$ and k be an integer for which a Regular Hadamard Matrix of order k exists. Then, there is a randomized construction of k orthogonal vectors $r_1, \ldots r_n \in \{-1, 1\}^d$, such that $\texttt{sign}(g)^\top r_i = d \cdot k^{-1/2}$. Moreover, $\mathbb{E}[g^\top r_i] = k^{-1/2} \cdot \|g\|_1$.*

Proof. We construct k orthogonal vectors $r_1, \ldots, r_k \in \{-1, 1\}^d$, where r_i is obtained by repeating the ith row of H_k d/k times (for simplicity, we assume that k divides d. Otherwise we pad r_i with zeros). We then multiply each r_i component-wise with $\texttt{sign}(g)$. By construction, the k vectors $r_i \in \{-1, 1\}^d$ are mutually orthogonal, and we have $\texttt{sign}(g)^\top r_i = \frac{d}{k} \cdot \sqrt{k} = d \cdot k^{-1/2}$, which is tight according to Lemma 6.

As the weight of the gradient g may not be uniformly distributed among its d components, we apply our construction to a random permutation of the signed gradient. We then obtain

$$\mathbb{E}[g^\top r_i] = \mathbb{E}\left[\sum_{j=1}^{d} |g^{(j)}| \cdot \texttt{sign}(g^{(j)}) \cdot r_i^{(j)}\right] \tag{13}$$

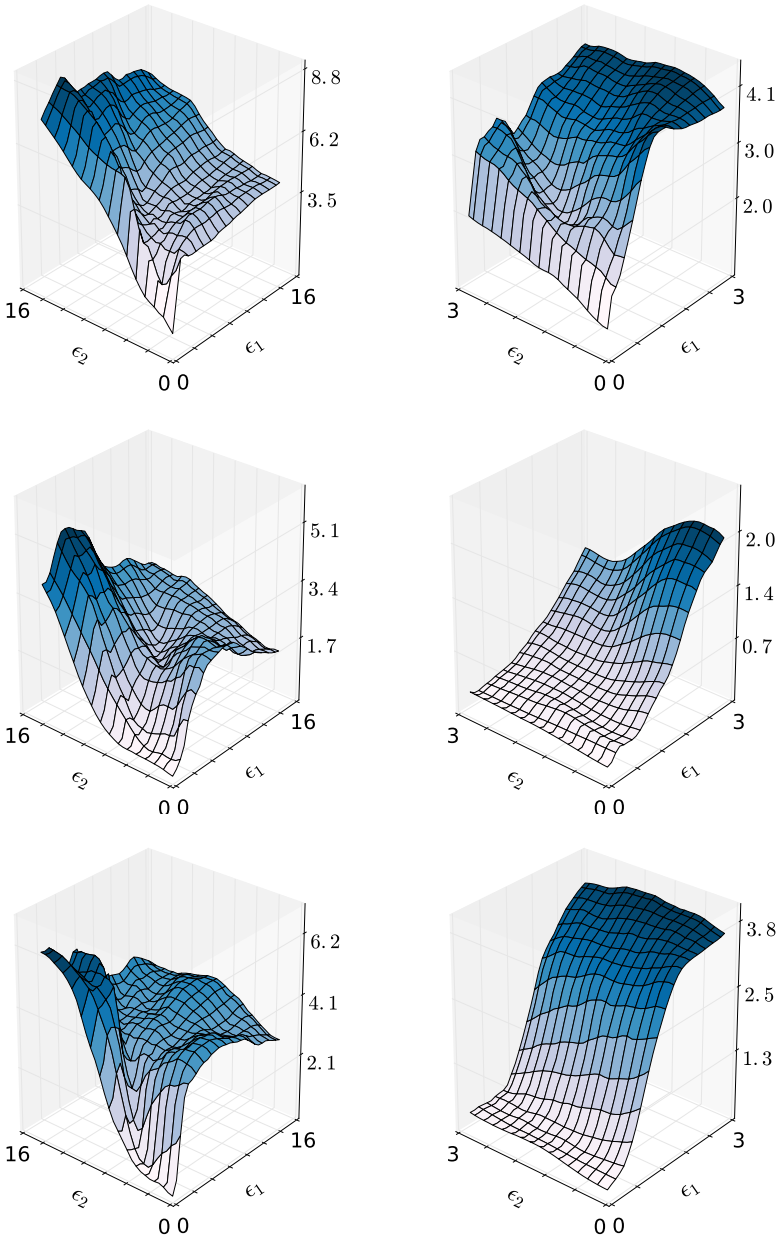
$$= \sum_{j=1}^{d} |g^{(j)}| \cdot \mathbb{E}\left[\texttt{sign}(g^{(j)}) \cdot r_i^{(j)}\right] = k^{-1/2} \cdot \|g\|_1 \,. \tag{14}$$

□

It can be shown that the bound in Lemma 7 can be attained if and only if the r_i are constructed from the rows of a Regular Hadamard Matrix (Colbourn, 2010). For general integers k for which no such matrix exists, other combinatorial designs may be useful for achieving looser bounds.

F ILLUSTRATIONS OF GRADIENT MASKING IN ADVERSARIAL TRAINING

In Section 3.3, we show that adversarial training introduces spurious curvature artifacts in the model's loss function around data points. As a result, one-shot attack strategies based on first-order approximations of the model loss produce perturbations that are non-adversarial. In Figures 4 and 5 we show further illustrations of this phenomenon for the Inception $v3_{adv}$ model trained on ImageNet by Kurakin et al. (2017b) as well as for the model A_{adv} we trained on MNIST.



(a) Loss surface for model $v3_{adv}$. (b) Zoom in of the loss for small ϵ_1, ϵ_2.

Figure 4: **Additional illustrations of the local curvature artifacts introduced by adversarial training on ImageNet.** We plot the loss of model $v3_{adv}$ on samples of the form $x^* = x + \epsilon_1 \cdot g + \epsilon_2 \cdot g^{\perp}$, where g is the signed gradient of $v3_{adv}$ and $g^{\perp}$ is an orthogonal adversarial direction, obtained from an Inception v4 model. The right-side plots are zoomed in versions of the left-side plots.